WORKED SOLUTION

STEP I 1994, Question 4

Pure • Inverse Tangent Integrals

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q4

The key idea

Part (i) is a half-angle identity, and part (ii) is the standard "complete the square, then use $\tan^{-1}$" integral. The final integral is part (ii) with $k = \cos\alpha$, which makes $\sqrt{1-k^2} = \sin\alpha$ and cancels the factor $\sin\alpha$ on top. The only real care needed is at the end: evaluating $\tan^{-1}$ of a tangent gives back the angle only when that angle lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Solution

Part (i)

Using the double-angle formulae $1 - \cos\alpha = 2\sin^2\frac{1}{2}\alpha$ and $\sin\alpha = 2\sin\frac{1}{2}\alpha\cos\frac{1}{2}\alpha$,

$$\frac{1-\cos\alpha}{\sin\alpha} = \frac{2\sin^2\frac{1}{2}\alpha}{2\sin\frac{1}{2}\alpha\cos\frac{1}{2}\alpha} = \frac{\sin\frac{1}{2}\alpha}{\cos\frac{1}{2}\alpha} = \tan\tfrac{1}{2}\alpha,$$

which holds whenever $\sin\alpha \neq 0$ (so both sides are defined). This is what we had to show.

Part (ii)

Complete the square in the denominator:

$$1 - 2kx + x^2 = (x-k)^2 + (1-k^2).$$

Since $|k| < 1$, the constant $1-k^2$ is positive, so we may write $1-k^2 = a^2$ with $a = \sqrt{1-k^2} > 0$. (This is exactly where the condition $|k| < 1$ is used: otherwise the denominator would factorise and we would get logarithms instead.) Substituting

$x - k = a\tan\theta$, so that $\mathrm{d}x = a\sec^2\theta\,\mathrm{d}\theta$,

$$\int \frac{\mathrm{d}x}{(x-k)^2 + a^2} = \int \frac{a\sec^2\theta}{a^2\tan^2\theta + a^2}\,\mathrm{d}\theta = \int \frac{1}{a}\,\mathrm{d}\theta = \frac{\theta}{a} + C$$
$$= \frac{1}{\sqrt{1-k^2}}\tan^{-1}\left(\frac{x-k}{\sqrt{1-k^2}}\right) + C,$$

as required. (Equivalently, differentiate the right-hand side: the chain rule gives $\frac{1}{a}\cdot\frac{1/a}{1+(x-k)^2/a^2} = \frac{1}{(x-k)^2+a^2}$.)

The final integral

Take $k = \cos\alpha$. Since $0 < \alpha < \pi$, we have $|k| < 1$, so part (ii) applies, and $\sin\alpha > 0$ gives

$$\sqrt{1-k^2} = \sqrt{\sin^2\alpha} = \sin\alpha$$

(not $-\sin\alpha$, because $\sin\alpha > 0$ on this interval). The factor $\sin\alpha$ in the numerator therefore cancels the factor $1/\sqrt{1-k^2}$ from part (ii):

$$\int_0^1 \frac{\sin\alpha}{1 - 2x\cos\alpha + x^2}\,\mathrm{d}x = \left[\tan^{-1}\left(\frac{x-\cos\alpha}{\sin\alpha}\right)\right]_0^1$$
$$= \tan^{-1}\left(\frac{1-\cos\alpha}{\sin\alpha}\right) - \tan^{-1}\left(\frac{-\cos\alpha}{\sin\alpha}\right)$$
$$= \tan^{-1}(\tan\tfrac{1}{2}\alpha) + \tan^{-1}(\cot\alpha),$$

using part (i) for the first term and the fact that $\tan^{-1}$ is an odd function for the second.

? **Ask yourself** Is $\tan^{-1}(\tan\theta)$ always equal to θ?

No: $\tan^{-1}$ takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\tan^{-1}(\tan\theta) = \theta$ only for θ in that interval. We check each term.

- Since $0 < \alpha < \pi$, we have $0 < \frac{1}{2}\alpha < \frac{\pi}{2}$, so $\tan^{-1}(\tan\frac{1}{2}\alpha) = \frac{1}{2}\alpha$.
- Write $\cot\alpha = \tan(\frac{\pi}{2} - \alpha)$. Since $0 < \alpha < \pi$, the angle $\frac{\pi}{2} - \alpha$ lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\tan^{-1}(\cot\alpha) = \frac{\pi}{2} - \alpha$.

Adding,

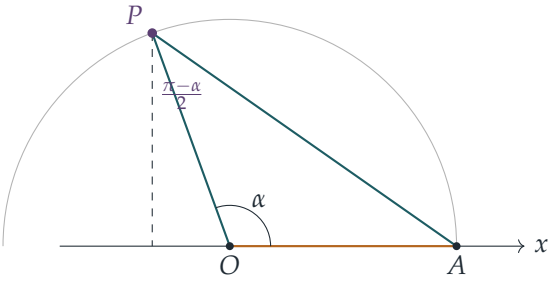
$$\int_0^1 \frac{\sin\alpha}{1 - 2x\cos\alpha + x^2}\,\mathrm{d}x = \frac{\alpha}{2} + \frac{\pi}{2} - \alpha = \frac{\pi - \alpha}{2},$$

which is the required result.

Otherwise: a picture

There is a lovely geometric reason for the answer. Let $P = (\cos\alpha, \sin\alpha)$ on the unit circle, O the origin and $A = (1, 0)$. As a point $X = (x, 0)$ moves along the x-axis, the angle θ between PX and the downward vertical through P satisfies $\tan\theta = (x - \cos\alpha)/\sin\alpha$,

and the integrand is exactly $\mathrm{d}\theta/\mathrm{d}x$. So the integral is the angle swept out as X moves from O to A, namely the angle OPA. Triangle OPA is isosceles with $OP = OA = 1$ and apex angle α at O, so its base angles are each $\frac{1}{2}(\pi - \alpha)$.



Checks

- At $\alpha = \frac{\pi}{2}$ the integral is $\int_0^1 \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}$, and the formula gives $\frac{1}{2}(\pi - \frac{\pi}{2}) = \frac{\pi}{4}$.
- Numerically, with $\alpha = 1$, Simpson's rule gives $1.0707963\ldots$, matching $\frac{1}{2}(\pi - 1) = 1.0707963\ldots$.
- As $\alpha \to \pi^-$ the integrand tends to 0 for each x in $[0, 1]$, and so does $\frac{1}{2}(\pi - \alpha)$.

Common slips

- Writing $\tan^{-1}(\cot\alpha) = \frac{\pi}{2} - \alpha$ without checking that $\frac{\pi}{2} - \alpha$ lies in the range of $\tan^{-1}$; the same move fails for α outside $(0, \pi)$.
- Taking $\sqrt{1 - \cos^2\alpha} = |\sin\alpha|$ to be $\sin\alpha$ without saying why; here it is because $0 < \alpha < \pi$.

Beyond STEP

As $\alpha \to 0^+$ the answer tends to $\frac{\pi}{2}$, even though at $\alpha = 0$ the integrand is identically zero: the integrand becomes a tall, narrow spike near $x = 1$ whose area does not shrink. This is the Poisson kernel for the half-plane, and swapping limits and integrals carelessly gives the wrong answer. When such a swap is allowed is a central question of first-year analysis; see Abbott, *Understanding Analysis*. The geometric "angle subtended" picture is developed beautifully in Needham, *Visual Complex Analysis*.

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